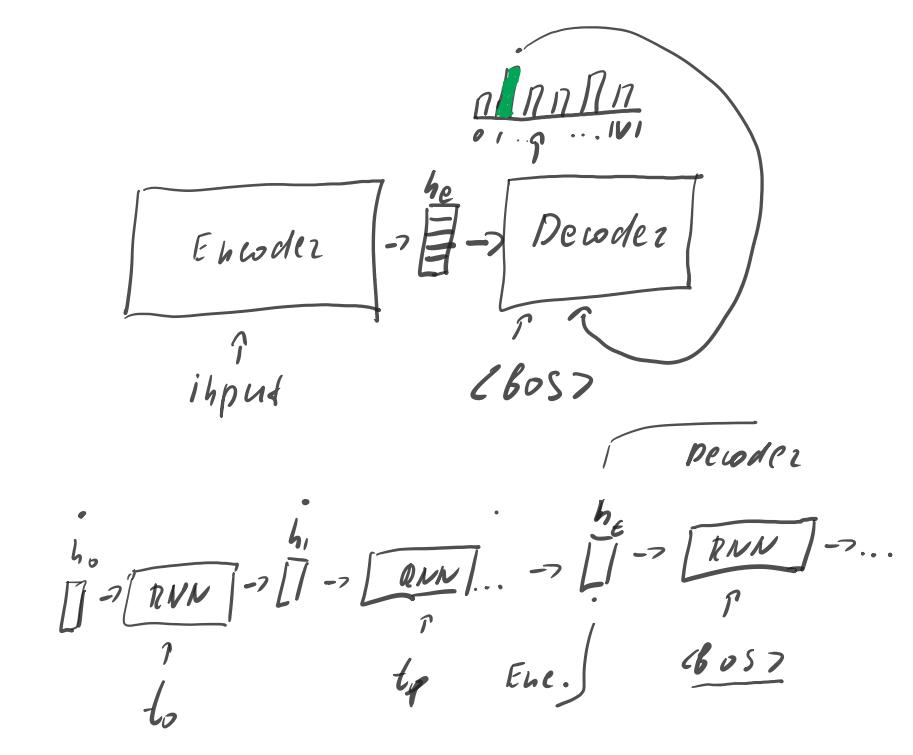
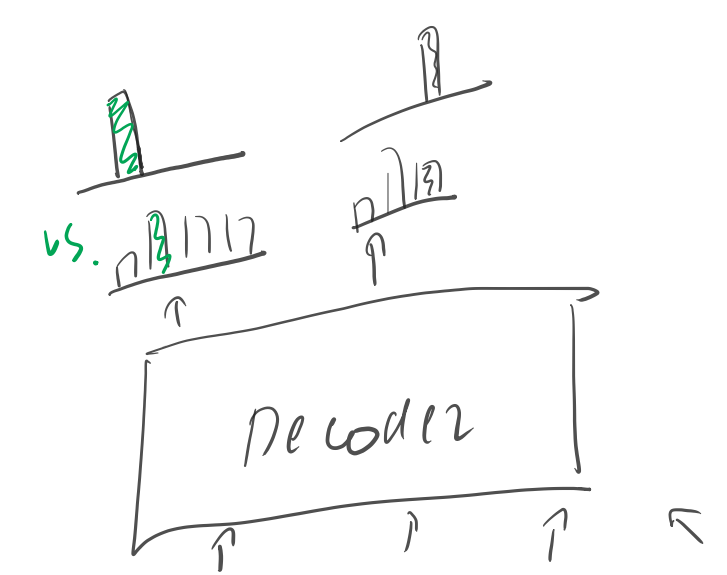
• Recap on Attention


0 1 … q … |V|
Encoder
h_e
Decoder
input
<bos>
Decoder
h_0
RNN
h_1
RNN
h_e
RNN
t_0
t_q
Enc.
<bos>



seq₀ <bos> кот на... мече
seq₁ <bos> песень на диване

• Tokenization

input: Deep␣Learning␣Engineer

[Deep, Learning, Engineer]

- Byte-Pair Encoding (BPE)

$V = [d, e, p, l, r, n, i, g, ee, in, lear, ing \ldots]$

de - 1
ee - 2
ep - 1
…
in - 2

- Byte BPE

utf-16 a - U+0145

$V = [0, 1, \ldots, A, B, C, D, a, b, c, d \ldots]$

utf-16 : d = U+0124 = U+0000 0001 0111 1011

кол-во бит

#8 utf-8

#16

- Word Piece

0. de - 0,015
ee - 0,356
ep - …
…
in - 0,333

$$\boxed{\text{score} = \frac{P(t_1, t_2)}{P(t_1)P(t_2)}}$$

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$$in: \frac{p(i,n)}{p(i)\,p(n)} \approx \frac{\#(i,n)}{\#(i)\,\#(n)} = \frac{2}{8}$$

$\#(n,i) \neq \#(i,n)$

- SentencePiece

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$V = [d, e, \ldots, ␣, \_lear, \_eng \ldots]$

• QKV attention

$$\begin{matrix} x_0 \\ x_1 \\ x_2 \end{matrix} \to X \begin{bmatrix} - x_0 - \\ - x_1 - \\ - x_2 - \end{bmatrix} \to X$$

seq_len × d

batch_size × seq_len × d

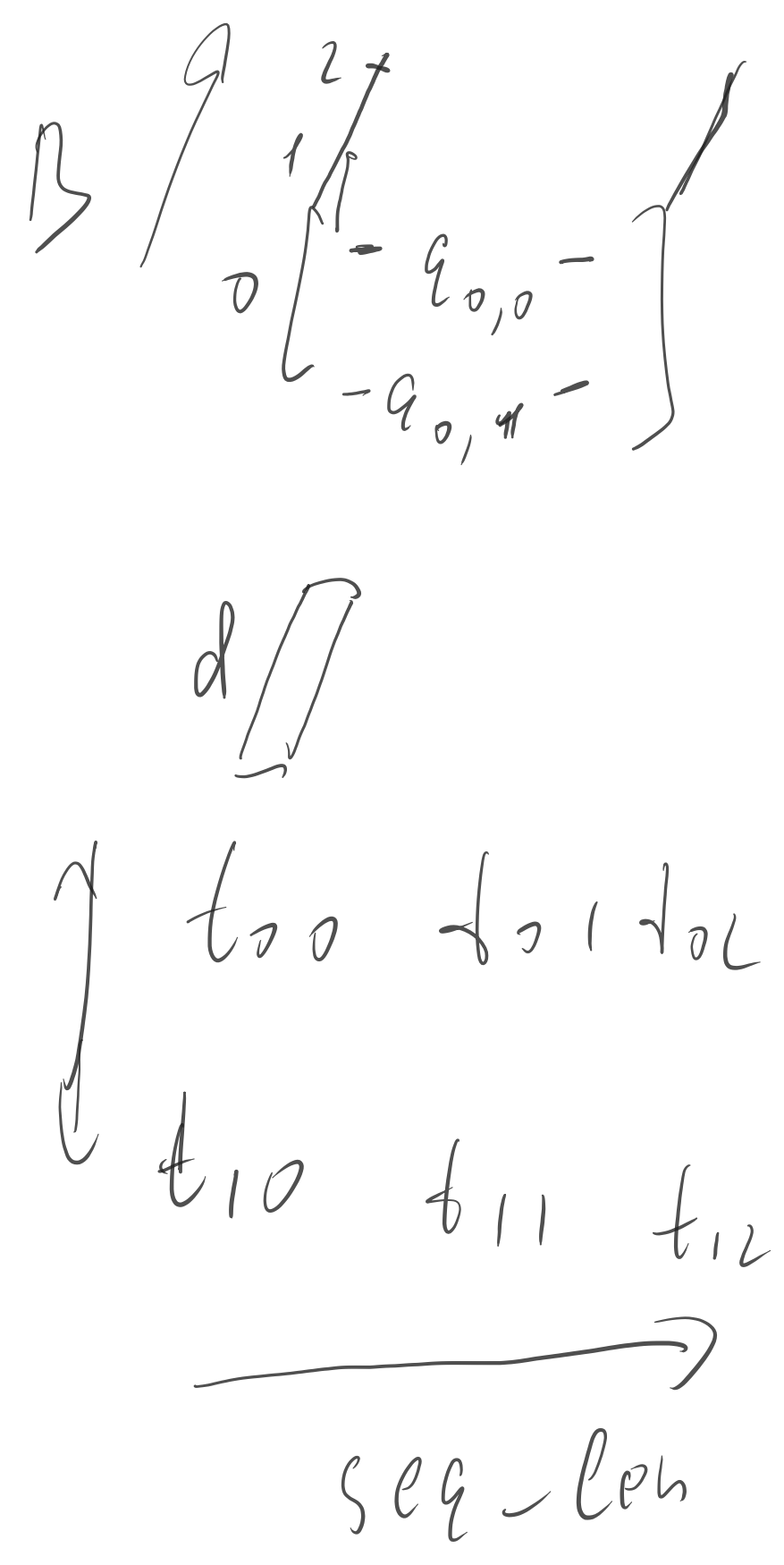
$$X \underset{d \times d'}{W^q} = Q = \begin{bmatrix} - q_0 - \\ - q_1 - \\ \ldots \end{bmatrix}$$

seq_len × d'

$X W^K = K$

$X W^V = V$

$$\text{Self-attn} = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V$$


B
q
2
1
0
$q_{0,0}$
$q_{0,d}$
d
t_{00} t_{01} t_{02}
t_{10} t_{11} t_{12}
seq_len

$$\Leftarrow \quad Q = \begin{bmatrix} - q_0 - \\ - q_1 - \\ \ldots \end{bmatrix}$$

$$\underset{d' \times seq\_len}{K^T} = \begin{bmatrix} | & | \\ k_0 & k_1 \\ | & | \end{bmatrix}$$

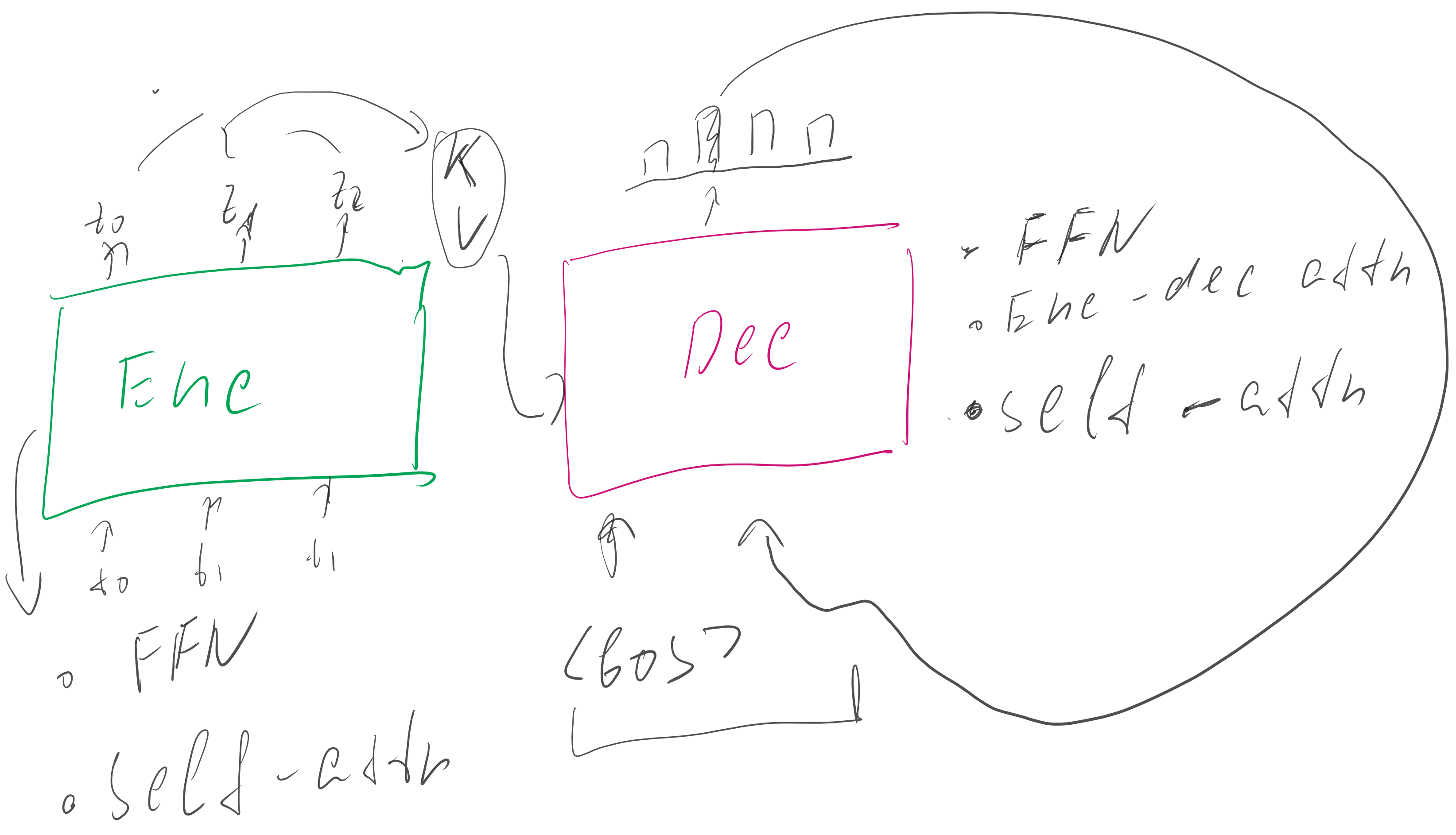
$$V = \begin{bmatrix} - v_0 - \\ - v_1 - \end{bmatrix}$$

$$\underset{seq\_len \times seq\_len}{QK^T} = \begin{bmatrix} \boxed{\langle q_0, k_0 \rangle \quad \langle q_0, k_1 \rangle} & \langle q_0, k_2 \rangle \ldots \\ \langle q_1, k_0 \rangle \quad \langle q_1, k_1 \rangle & \ldots \end{bmatrix} = \begin{bmatrix} \ldots \langle q_i, k_j \rangle \ldots \end{bmatrix}$$

$$\text{softmax}(QK^T) = \begin{bmatrix} - \text{оценки } q_0 \text{ к } k_j - \\ - \text{оценки } q_1 \text{ к } k_j - \\ \ldots \end{bmatrix}$$

$$\text{softmax}(QK^T) \begin{bmatrix} - v_0 - \\ - v_q - \end{bmatrix}$$

• Masked attn.


K
V
t_0 t_1 t_2
Enc
t_0 t_1 t_1
◦ FFN
◦ self-attn
Dec
<bos>
• FFN
◦ Enc-dec attn
◦ self-attn

$$\text{softmax}\left(QK^T + \begin{bmatrix} 0 & -\text{inf} & -\text{inf} \\ 0 & 0 & -\text{inf} \\ 0 & 0 & 0 \end{bmatrix}\right)$$

seq_len × seq_len